Pump It Up

Your heart and blood vessels (circulatory system) and your lungs and airways (respiratory system) work as one team: the lungs bring in oxygen and get rid of carbon dioxide, and the blood carries these gases to and from every cell.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 16	Your Body's Delivery Team
Lesson 2	Fri Nov 20	Inside the Heart
Lesson 3	Fri Jan 15	Blood and Breathing Models
Lesson 4	Fri Feb 12	Mock Tournament
Lab I1	Before Lesson 3	Airway Map and Breathing Lab (home or an open Friday)
Lab I2	Before Lesson 4	Case Files: Heart and Blood Conditions (home or an open Friday)
Lab I3	Before Lesson 4	Case Files: Lung Conditions and Healthy Habits (home or an open Friday)

At the tournament

Pairs rotate through timed hands-on stations for up to 50 minutes (label diagrams, study models, answer scenario questions about the heart, blood, lungs, and diseases) and bring only pencils, no notes.

Keep every handout for this event behind this divider, in order. Bring this binder to every Diagram & Decode Fridays meeting.

Pump It Up: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can name the 4 heart chambers (right and left atria, right and left ventricles) and the aorta, and say what each does.	L2 L4	☐	☐
I can name the 4 valves in order (tricuspid, pulmonary, mitral, aortic) and explain that valves are one-way doors.	L2 L4	☐	☐
I can trace blood from the body, through the heart and lungs, and out the aorta.	L2	☐	☐
I can name the parts of blood (red cells, white cells, platelets, plasma) and their jobs.	L3 L4	☐	☐
I can compare arteries, veins, and capillaries.	L3 L4	☐	☐
I can put the airway in order: nose (or mouth), pharynx, larynx, trachea, bronchi, bronchioles, alveoli.	L3 L4 I1	☐	☐
I can explain the lungs, lobes, and diaphragm, and use a model to show breathing.	L3 L4 I1	☐	☐
I can explain how oxygen and carbon dioxide trade places in the lungs and how the two systems work together.	L1 L3 L4 I1	☐	☐
I can measure pulse and breathing rate and explain why both go up with exercise.	L1 I1	☐	☐
I can explain sickle cell anemia, heart attack, and stroke (organ affected, causes, prevention, treatment).	L4 I2	☐	☐
I can explain asthma and cystic fibrosis (organ affected, causes, prevention, treatment).	L4 I3	☐	☐
I can explain COPD, lung cancer, and congestive heart failure (organ affected, causes, prevention, treatment).	L4 I2 I3	☐	☐
I can tell which conditions are genetic and which come mostly from habits or the environment, and name healthy habits.	I2 I3	☐	☐
I can answer station questions on diagrams, models, and scenarios with no notes.	L4	☐	☐

Your Body's Delivery Team

TODAY I CAN

- I can explain how oxygen and carbon dioxide trade places in the lungs and how the two systems work together.
- I can measure pulse and breathing rate and explain why both go up with exercise.

Every cell in your body needs oxygen, a gas in the air. Your lungs bring oxygen in. Your heart and blood deliver it. On the way back, blood picks up carbon dioxide, a waste gas, and your lungs breathe it out.

Today's steps: run it yourselves (22 minutes)

You need: sample station card, bendable straws (optional). Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Read the Key words box aloud. Find your pulse: two fingers (not your thumb) on the thumb side of your wrist or the side of your neck. Each beat is your heart squeezing blood out.
2. Event Judge 6 min. Everyone sits still. Time 15 seconds while everyone counts pulse beats, then times 4. Time 30 seconds while partners count each other's breaths, then times 2. Write both in the At rest row of My data.
3. Team 2 min. March in place fast or do jumping jacks for 1 minute. Anyone can sit out and be the timekeeper.
4. Event Judge 5 min. Right away, time the pulse and breath counts again for the Right after exercise row. Rest 1 minute, then count once more for the last row.
5. Both 4 min. Answer the Think about it questions. Hint: what do working muscles need more of? Then read Follow the air aloud and point to each part on your own body.
6. Event Captain 2 min. Read the sample station card aloud; each pair writes its answer on the whiteboard.

Check your work: Ask your Event Captain for the Lesson 1 answer card. Compare your numbers to the usual ranges and check your Think about it answers. Fix any you missed in a different color.

Key words

Circulatory system	Heart, blood, and blood vessels. It moves blood around the body.
Respiratory system	Nose, airways, and lungs. It brings air in and out.
Oxygen	The gas your cells need. You breathe it in.
Carbon dioxide	The waste gas your cells make. You breathe it out.
Pulse	The beat you feel in an artery each time the heart pumps.

My data

	Pulse (beats in 15 sec x 4)	Breaths (in 30 sec x 2)
At rest		
Right after exercise		
After resting again		

Think about it

1. What happened to your pulse and breathing after exercise?

2. Why do both go up at the same time?

3. Which gas do you breathe in that your body needs? Which gas do you breathe out?

Follow the air (you will learn each part later)

Nose (or mouth) > pharynx (throat) > larynx (voice box) > trachea (windpipe) > bronchi (one tube to each lung) > bronchioles (small branches) > alveoli (tiny air sacs where oxygen goes into the blood).

At home before next time: Family pulse check (15 min). Teach someone at home to find a pulse. Measure it at rest and after 1 minute of marching; write both numbers on the back. Make 5 flashcards (circulatory system, respiratory system, oxygen, carbon dioxide, pulse) and quiz yourself twice. Case File: record everyone's numbers and graph them in Case #1: Family Pulse Check (p. 8) for the Pulse Detective badge.

Inside the Heart

TODAY I CAN

- I can name the 4 heart chambers (right and left atria, right and left ventricles) and the aorta, and say what each does.
- I can name the 4 valves in order (tricuspid, pulmonary, mitral, aortic) and explain that valves are one-way doors.
- I can trace blood from the body, through the heart and lungs, and out the aorta.

Your heart is a strong muscle that pumps blood about 100,000 times a day. It has 4 chambers, 4 valves, and 2 sides that work at the same time.

Today's steps: run it yourselves (22 minutes)

You need: floor heart diagram (rolled up), 10 red oxygen cards, 10 blue carbon dioxide cards, sticky notes. Set the timer for each step.

1. Both 5 min. Set the timer for 5 minutes. Unroll the heart diagram on the floor and read the Heart parts box aloud. The heart's right side is on YOUR left. Put sticky-note labels on the 4 chambers: atria on top, ventricles on the bottom.
2. Both 5 min. Label the 4 valves with sticky notes, using the Heart parts box. Memory tip: 'Try Pulling My Aorta' = Tricuspid, Pulmonary, Mitral, Aortic.
3. Team 8 min. Be the blood. Put blue cards at Body and red cards at Lungs. One person walks the path holding a blue card and names each part; trade blue for red at the Lungs and red for blue at the Body. Others check with the handout, and everyone takes a turn.
4. Both 4 min. Fill in the Path of blood blanks and answer Check yourself. If time is left, draw the 4-box heart.

Check your work: Ask your Event Captain for the Lesson 2 answer card. Check your Path of blood blanks and Check yourself answers. Fix any you missed in a different color.

Heart parts

Atria (atrium = one)	The 2 top chambers. They receive blood coming into the heart.
Ventricles	The 2 bottom chambers. They pump blood out of the heart.
Tricuspid valve	Between the right atrium and right ventricle.
Pulmonary valve	Between the right ventricle and the artery to the lungs.
Mitral valve	Between the left atrium and left ventricle.
Aortic valve	Between the left ventricle and the aorta.
Aorta	The biggest artery. It carries oxygen-rich blood from the left ventricle to the body.

Path of blood

Body > _____ > _____ valve > _____ > _____ valve > _____ (drop off carbon dioxide, pick up oxygen) > _____ > _____ valve > _____ > _____ valve > _____ > Body

Draw a 4-box heart. Label the chambers. Remember: the heart's right side is on YOUR left.

Check yourself

1. Which side of the heart has oxygen-poor blood? Oxygen-rich blood?

---

2. What do valves do?

---

3. Which chamber has the thickest wall? Why?

At home before next time: Do Independent Lab I1 (Airway Map and Breathing Lab) before Lesson 3. Also make 7 heart flashcards (atrium, ventricle, tricuspid, pulmonary valve, mitral, aortic valve, aorta) and quiz yourself until all 7 are right twice. Case File: label and color Case #2: The Heart Map (p. 9) for the Heart Mapper and Blood Tracer badges.

Blood and Breathing Models

TODAY I CAN

- I can name the parts of blood (red cells, white cells, platelets, plasma) and their jobs.
- I can compare arteries, veins, and capillaries.
- I can put the airway in order: nose (or mouth), pharynx, larynx, trachea, bronchi, bronchioles, alveoli.
- I can explain the lungs, lobes, and diaphragm, and use a model to show breathing.
- I can explain how oxygen and carbon dioxide trade places in the lungs and how the two systems work together.

Blood is a delivery service that travels through tubes called blood vessels. Your lungs load it with oxygen. Today you build models of both.

Today's steps: run it yourselves (27 minutes)

You need: blood in a jar kit for each pair (clear cup, corn syrup, red candies, white beans, sprinkles, spoon), piece of garden hose, floppy straw, thread, 2 balloon lung kits (cut bottle, small balloon, bendable straw, clay, cut large balloon, rubber band), grape bunch picture, paper towels. Set the timer for each step.

1. Both 7 min. Set the timer for 7 minutes. Blood in a jar: pour in corn syrup (plasma, the liquid part that carries food, water, and waste). Add red candies (red blood cells, carry oxygen), white beans (white blood cells, fight germs), and sprinkles (platelets, help a cut stop bleeding). Fill in the Parts of blood table.
2. Both 5 min. Pass around the hose (artery), straw (vein), and thread (capillary). Arteries carry blood Away from the heart and have thick, stretchy walls; veins bring it back, with thinner walls and valves; capillaries are so tiny that gases and food trade places there. Fill in the Blood vessels table.
3. Both 10 min. Build the balloon lung: rubber-band the small balloon to the end of the straw, push the straw through the bottle neck, and seal it with clay. Stretch the cut balloon over the bottom. Pull the knot down, then push up. Fill in the Balloon lung model table and answer Check yourself 2.
4. Both 3 min. Look at the grape picture (stems = bronchioles, grapes = alveoli). Fill in the Gas exchange blanks and answer Check yourself 1 and 3.
5. Partner A 2 min. Throw the jar contents in the trash (candy is not for eating) and wipe up. Point to a part of the lung model; Partner B names the body part it stands for. Switch.

Check your work: Ask your Event Captain for the Lesson 3 answer card. Check both blood tables, the model table, the Gas exchange blanks, and Check yourself. Fix any you missed in a different color.

Parts of blood

Part	What it looks like	Job
Plasma		
Red blood cells		
White blood cells		
Platelets		

Blood vessels

Vessel	Carries blood…	Walls and size
Artery		
Vein		
Capillary		

Balloon lung model

Model part	Body part it stands for
Bottle	
Straw	
Small balloon	
Bottom balloon	

Gas exchange

In the alveoli, _____ moves from the air into the blood. _____ moves from the blood into the air, and you breathe it out. The blood then carries oxygen to the heart, which pumps it to the whole body. The two systems work as a team.

Check yourself

1. In which vessels do oxygen and carbon dioxide trade places with the cells?

---

2. What happens when the diaphragm tightens and moves down?

---

3. Which vessels carry blood away from the heart? Which bring it back?

At home before next time: Do Independent Labs I2 and I3 (the Case Files) before Lesson 4, the mock tournament. Also make 7 flashcards (plasma, red blood cells, white blood cells, platelets, artery, vein, capillary) and quiz yourself twice. Case File: find 5 body models at home in Case #3: The Model Hunt (p. 10) for the Model Maker badge.

Mock Tournament

TODAY I CAN

- I can name the 4 heart chambers (right and left atria, right and left ventricles) and the aorta, and say what each does.
- I can name the 4 valves in order (tricuspid, pulmonary, mitral, aortic) and explain that valves are one-way doors.
- I can name the parts of blood (red cells, white cells, platelets, plasma) and their jobs.
- I can compare arteries, veins, and capillaries.
- I can put the airway in order: nose (or mouth), pharynx, larynx, trachea, bronchi, bronchioles, alveoli.
- I can explain the lungs, lobes, and diaphragm, and use a model to show breathing.
- I can explain how oxygen and carbon dioxide trade places in the lungs and how the two systems work together.
- I can explain sickle cell anemia, heart attack, and stroke (organ affected, causes, prevention, treatment).
- I can explain asthma and cystic fibrosis (organ affected, causes, prevention, treatment).
- I can explain COPD, lung cancer, and congestive heart failure (organ affected, causes, prevention, treatment).
- I can answer station questions on diagrams, models, and scenarios with no notes.

Mock tournament answer sheet. No notes. Work with your partner. When time is called, stop and move.

Today's steps: run it yourselves (27 minutes)

You need: 5 station cards (A to E), Station A: floor heart diagram with 8 number labels, Station B: sealed blood jar, Station C: balloon lung model, Station E: case cards 2, 4, 7, 8. Set the timer for each step.

1. Event Judge 1 min. Read the rules aloud: stay at your station until time is called, no going back, no notes, whisper with your partner only. One pair starts at Station A, the other at B.
2. Event Judge 4 min. Set the timer for 4 minutes. Pairs work their first station, then the Judge calls 'Rotate!' Tip: find the aorta first on the heart diagram.
3. Both 4 min. Second station. Move forward one letter when time is called.
4. Both 4 min. Third station.
5. Both 4 min. Fourth station.
6. Both 4 min. Fifth station. Skip any question you are stuck on and come back if time is left.
7. Event Captain 6 min. Read the answers from the answer card while each pair marks its own sheet. Each pair circles its 2 weakest topics and writes them in the at-home box.

Check your work: Your Event Captain reads the Lesson 4 answer card aloud. Mark your own sheet and fix any you missed in a different color. Your 2 weakest topics are your study list.

Station A: Heart diagram (and: which valve does blood pass through to leave for the body? _____)

#	Part	#	Part
1		5	
2		6	
3		7	
4		8	

Station B: Blood model

1. What does each item stand for? Syrup _____ Red candies _____ White beans _____ Sprinkles _____

2. Which part carries oxygen? _____ Helps a cut stop bleeding? _____ Is the liquid? _____

Station C: Breathing model

1. What body part does each part of the model stand for?

2. Pull the bottom balloon down. What happens? _____

3. Put the air path in order starting at the nose.

Station D: Working together

Question	Answer
1. Which gas does blood pick up in the lungs?	
2. Where do the gases trade places?	
3. Which waste gas leaves the blood in the lungs?	
4. In a race, do pulse and breathing go up or down?	
5. Which vessels carry blood back to the heart?	

Station E: Case files

Card	Condition	Organ affected	One cause or treatment
2			
4			
7			
8			

At home before next time: Final review (20 min). My 2 weakest topics: _____ and _____. Quiz those cards until you get them right 3 times, then have someone quiz you on 10 random cards. Draw the heart path once more from memory. Case File: solve Case #4: Mystery Patients (p. 11) for the Case Solver badge.

Lab I1: Airway Map and Breathing Lab

THIS LAB HELPS ME

- I can put the airway in order: nose (or mouth), pharynx, larynx, trachea, bronchi, bronchioles, alveoli.
- I can explain the lungs, lobes, and diaphragm, and use a model to show breathing.
- I can explain how oxygen and carbon dioxide trade places in the lungs and how the two systems work together.
- I can measure pulse and breathing rate and explain why both go up with exercise.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Your respiratory system brings fresh air deep into your lungs, where oxygen goes into the blood. In this lab you build an airway reference, measure how much your chest grows when you breathe in, and collect breathing data like a scientist.

Steps (30 minutes)

You need: this handout, 10 index cards, string or yarn (about 1 meter), ruler or tape measure, a timer. Set the timer for each step.

1. Me 5 min. Set a timer for 5 minutes. Read Parts of the respiratory system aloud. Point to each part on your own body. Hum and feel your larynx buzz.
2. Me 6 min. Make 10 flashcards: nose and nasal cavity, mouth, pharynx, larynx, trachea, bronchi, bronchioles, alveoli, lungs and lobes, diaphragm. Name on the front, job on the back.
3. Me 3 min. Shuffle the 8 airway cards (nose to alveoli). Lay them in order with no peeking, then check with the box. Do it twice.
4. Me 5 min. Chest measure: wrap the string around your chest under your arms. Breathe all the way out, pinch the string where it meets, and measure it with the ruler. Then take a big breath in and measure again. Fill in the Chest measure table.
5. Me 8 min. Breathing and pulse data: for each row, count breaths for 30 seconds (times 2) and pulse for 15 seconds (times 4). Rest 1 minute between rows. March in place only as hard as feels okay.
6. Me 3 min. Answer the Think about it questions.

Check your work: Check your airway order against the Parts box. On an open Friday or at Lesson 3, ask for the Lab I1 answer card from the Pump It Up bin and fix any you missed in a different color.

Parts of the respiratory system

Nose and nasal cavity	Air comes in. Hairs and mucus trap dust; the air gets warm and wet.
Mouth	Another way for air to get in and out.
Pharynx	The throat. Air and food both pass through it.
Larynx	The voice box. Air passing through it lets you talk.
Trachea	The windpipe. Stiff rings keep it open.
Bronchi	Two tubes that split off the trachea, one into each lung.
Bronchioles	Smaller and smaller tubes branching inside the lungs.
Alveoli	Millions of tiny air sacs covered with capillaries. Gases trade places here.
Lungs and lobes	Two spongy organs. Right lung: 3 lobes. Left lung: 2 lobes.
Diaphragm	A dome-shaped muscle under the lungs. Down = breathe in. Up = breathe out.

Chest measure

	Chest size (cm)
After breathing all the way out	
After a big breath in	
Difference (in minus out)	

Breathing and pulse data

Activity	Breaths in 30 sec x 2	Pulse in 15 sec x 4
Lying down (1 minute first)		
Sitting		
Standing		
Right after 1 minute of marching in place		

Think about it

1. Which activity had the most breaths and the fastest pulse? Why does your body do that?

2. Your chest got bigger when you breathed in. Which muscle moved down to make room?

3. Why does the left lung have only 2 lobes?

4. Where do oxygen and carbon dioxide trade places in the lungs?

Bring to the next meeting: Bring this handout with your data tables and your 10 airway flashcards.

Lab I2: Case Files: Heart and Blood Conditions

THIS LAB HELPS ME

- I can explain sickle cell anemia, heart attack, and stroke (organ affected, causes, prevention, treatment).
- I can explain COPD, lung cancer, and congestive heart failure (organ affected, causes, prevention, treatment).
- I can tell which conditions are genetic and which come mostly from habits or the environment, and name healthy habits.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. At the tournament you may be asked about 8 conditions. For each one, know: what organ it affects, what causes it, how to prevent it, and how doctors treat it. This lab covers the 4 heart and blood conditions. Many are not anyone's fault, and doctors have good ways to help.

Steps (29 minutes)

You need: this handout, 4 index cards, a red colored pencil. Set the timer for each step.

1. Me 8 min. Set a timer for 8 minutes. Read the 4 condition boxes aloud. Then draw a normal red blood cell (a round disk) and a sickle cell (a curved crescent) in the Draw it box.
2. Me 5 min. Case files: read each case. Write which condition it is and the clue that told you.
3. Me 8 min. Cover the condition boxes with a sheet of paper. Fill in the Condition chart from memory. Then uncover the boxes, check, and fix mistakes in red.
4. Me 3 min. Answer the Check yourself questions.
5. Me 5 min. Make 4 flashcards (sickle cell anemia, heart attack, stroke, congestive heart failure). On the back write organ, cause, prevention, and treatment. Quiz yourself twice.

Check your work: Check your chart against the condition boxes. On an open Friday or at Lesson 4, ask for the Lab I2 answer card from the Pump It Up bin and fix any you missed in a different color.

Sickle cell anemia

Affects	Blood: red blood cells.
What happens	Red blood cells are shaped like a curved crescent (a sickle) instead of a round disk. They are stiff and sticky, can get stuck in small vessels, and do not carry as much oxygen. People can feel tired or have pain.
Cause	Genetic. A person gets the sickle cell gene from both parents. It is not catching.
Prevention	It cannot be prevented, but babies are tested at birth. Drinking lots of water, resting, and staying warm help prevent pain episodes.
Treatment	Medicines, blood transfusions, and for some people a bone marrow transplant.

Heart attack

Affects	The heart muscle and the arteries that feed it.
What happens	A fatty build-up (plaque) or a clot blocks an artery, so part of the heart muscle does not get oxygen.
Causes	Smoking, high blood pressure, high cholesterol, diabetes, not enough exercise, family history.
Prevention	Exercise, healthy food, no smoking, regular checkups.
Treatment	Call 911 fast. Doctors use medicine, a tiny tube (stent) to open the artery, or surgery.

Stroke

Affects	The brain and the blood vessels that feed it.
What happens	A clot blocks blood to part of the brain, or a vessel leaks. Signs: face drooping, arm weak, speech mixed up. Time to call 911.
Causes	High blood pressure, smoking, heart problems, diabetes. Sickle cell anemia can raise the risk in kids.
Prevention	Same healthy habits as for the heart; keep blood pressure healthy.
Treatment	Fast medicine to break up the clot or a procedure to remove it, then therapy to relearn skills.

Congestive heart failure

Affects	The heart. Fluid can back up into the lungs and legs.

What happens	The heart still beats, but it pumps weaker than it should. People get out of breath and their ankles swell.
Causes	A past heart attack, high blood pressure, or damaged valves.
Prevention	Healthy food, exercise, no smoking, keep blood pressure healthy.
Treatment	Medicines (including water pills to remove extra fluid), less salt, and sometimes devices or surgery.

Draw it: a normal red blood cell and a sickle cell

Case files

Case	Condition	Clue that told you
A. Maya's red blood cells are shaped like crescent moons. She gets tired and has pain some days. Her parents both carry a gene for it.		
B. Grandpa's face drooped on one side and his words were mixed up. The family called 911 right away.		
C. Mr. Park felt chest pain because an artery to his heart muscle was blocked. Doctors put in a stent.		
D. Aunt June's heart does not pump as strongly as it should, so fluid builds up and her ankles swell. She eats less salt and takes medicine.		

Condition chart (part 1)

Condition	Organ affected	Cause	Prevention	Treatment
Sickle cell anemia				
Heart attack				
Stroke				
Congestive heart failure				

Check yourself

1. Which of these 4 conditions is genetic?

2. How is a heart attack different from congestive heart failure?

3. Someone's face droops and their words are mixed up. What should you do?

Bring to the next meeting: Bring this handout with your chart filled in and your 4 condition flashcards.

Lab I3: Case Files: Lung Conditions and Healthy Habits

THIS LAB HELPS ME

- I can explain asthma and cystic fibrosis (organ affected, causes, prevention, treatment).
- I can explain COPD, lung cancer, and congestive heart failure (organ affected, causes, prevention, treatment).
- I can tell which conditions are genetic and which come mostly from habits or the environment, and name healthy habits.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. This lab covers the 4 lung conditions. Then you sort all 8 conditions and plan healthy habits that protect your heart and lungs.

Steps (29 minutes)

You need: this handout and your Lab I2 handout, 4 index cards, 2 drinking straws. Set the timer for each step.

1. Me 7 min. Set a timer for 7 minutes. Read the Asthma, Cystic fibrosis, COPD, and Lung cancer boxes aloud.
2. Me 2 min. Airway model: breathe out one gentle breath through a straw. Then pinch a second straw halfway flat and breathe out through it. Which was harder? Write it under Check yourself 1. Skip this step if you have asthma or feel dizzy.
3. Me 4 min. Case files: read each case. Write which condition it is and the clue that told you.
4. Me 6 min. Cover the condition boxes. Fill in the Condition chart from memory, then uncover, check, and fix mistakes in another color.
5. Me 4 min. Sort all 8 conditions (use your Lab I2 handout too) into the Genetic or habits and environment table.
6. Me 3 min. Answer the Healthy habits question.
7. Me 3 min. Make 4 quick flashcards (asthma, cystic fibrosis, COPD, lung cancer): name on the front, organ and one treatment on the back.

Check your work: Check your chart against the condition boxes. On an open Friday or at Lesson 4, ask for the Lab I3 answer card from the Pump It Up bin and fix any you missed in a different color.

Asthma

Affects	Lungs: the airways (bronchi and bronchioles).
What happens	Airways swell, the muscles around them tighten, and they make extra mucus. Breathing gets hard; there may be coughing or wheezing.
Causes	Often runs in families. Triggers include smoke, dust, pollen, pet dander, cold air, colds, and hard exercise.
Prevention	You cannot always prevent having asthma, but you can prevent attacks: avoid triggers and take controller medicine.
Treatment	Inhalers: a quick-relief (rescue) inhaler opens airways fast; a controller medicine lowers swelling.

Cystic fibrosis (CF)

Affects	Lungs and the digestive system (especially the pancreas).
What happens	The body makes thick, sticky mucus that clogs airways and traps germs, and makes it hard to digest food.
Cause	Genetic. A child gets a CF gene from both parents. It is not catching.
Prevention	It cannot be prevented. Babies are tested at birth so care can start early.
Treatment	Daily airway clearing (special vest or chest tapping), inhaled medicines, enzyme pills with meals, and newer medicines that help the body work better.

COPD (Chronic Obstructive Pulmonary Disease)

Affects	Lungs: the airways and alveoli get damaged over many years.
What happens	It is hard to breathe out all the way, so people feel out of breath and cough.
Causes	Mostly smoking. Also breathing smoke, dust, or polluted air for a long time.
Prevention	Do not smoke and stay away from smoke.
Treatment	Quit smoking, inhalers, extra oxygen, and breathing exercise programs.

Lung cancer

Affects	Lungs.
What happens	Some cells grow out of control and make a lump called a tumor.
Causes	Mostly smoking and secondhand smoke. Also radon (a gas from the ground), pollution, and sometimes genes.
Prevention	Do not smoke, avoid smoke, test your home for radon.
Treatment	Surgery, chemotherapy, radiation, and other medicines.

Case files

Case	Condition	Clue that told you
E. When Leo runs in cold air, he wheezes and his chest feels tight. His inhaler helps.		
F. Sam was born with thick, sticky mucus in his lungs and takes enzyme pills with meals.		
G. Mr. Green smoked for 40 years. Now it is hard to breathe out and he walks slowly; he uses oxygen.		
H. A scan found a lump of cells growing out of control in Ms. Diaz's lung. She has a radon test kit now for her home.		

Condition chart (part 2)

Condition	Organ affected	Cause	Prevention	Treatment
Asthma				
Cystic fibrosis				
COPD				
Lung cancer				

Genetic or habits and environment?

Genetic (born with it)	Mostly habits or environment	A mix

Check yourself

1. Airway model: which straw was harder to breathe out through? How is that like asthma?

Healthy habits

1. List 3 habits that protect your heart and lungs and one condition each one helps prevent.

Bring to the next meeting: Bring this handout and your Lab I2 handout with both charts and the sorting table filled in, plus your 4 lung flashcards.